

Learning to Optimize: Edge-Based Graph Neural Networks Trained on MILP-Optimized Routing Paths

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Problem

Modern networks experience rapidly changing traffic, requiring routing that adapts to congestion and link utilization. Mixed Integer Linear Programming (MILP) provides optimal routing decisions but is too computationally expensive for real-time use. We address this by learning MILP-optimized routing decision using an edge-based Graph Neural Network (GNN) that learns to optimize based on current network state.

Dataset

We study routing on the GEANT network [2] **G(23,74)** containing 11,460 traffic matrices, one every 15 minutes for four months. With some assumptions we generate a sequence of flows such that when combined they translate to the same TM [1]. And repeat this for all 11460 matrices for a sequence of flows (approx. 5 million), where a flow is defined by: **source node, destination node, bandwidth, duration, and arrival time.**



Models

MILP Routing Model: For each sequential flow arrival or departure, we solve a MILP that minimizes the maximum link utilization (MLU) and generate a unique path. The MILP enforces:

- Path continuity constraints,
- Link capacity constraints,
- Sequential update of active flows' contributions.

GNN Routing Model: A fixed-topology, edge-based GNN predicts the path used by each new flow.

- Node features: one-hot source and destination indicators.
- Edge features: current utilization ratio and flow bandwidth (negative for departures).
- Output: a 74-dimensional binary vector indicating whether each edge is used in the path.

References

- [1] Reis et al. *Deep Neural Networks for Network Routing*. IJCNN, 2019.
- [2] Uhlig et al. *Providing Public Intra-domain Traffic Matrices*. SIGCOMM CCR, 2006.

Acknowledgements

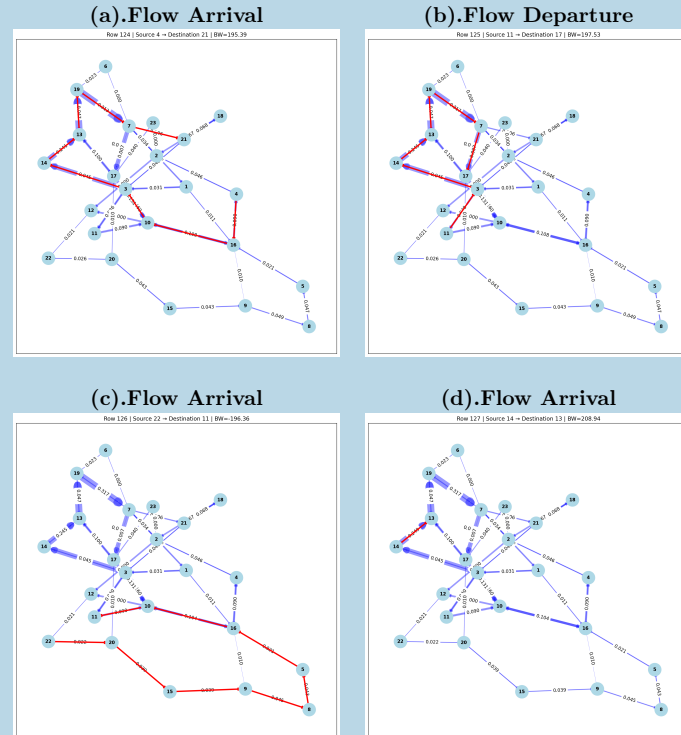
The GEANT traffic matrices were used for this study. The author acknowledges the computational resources provided by CDAC, Pune. We thank the CASML 2025 community for feedback.

Abstract

In this report, we demonstrate an edge-based GNN model capable of routing traffic in a network $G(23, 74)$ through optimized paths. Routing decisions are computed using a Mixed Integer Linear Programming (MILP) formulation that minimizes maximum link utilization (MLU) while considering the current network state. The optimal path between two nodes is used as supervised ground truth for GNN training. The study utilizes four months of real-world traffic data from the GEANT network to train the model. The proposed GNN achieves a mean absolute error of 0.0745 when predicting the MILP-optimized routing paths. This demonstrates that the model successfully learns to optimize the routing decisions in real time, offering a significant computational advantage over conventional optimization-based methods that are infeasible for runtime deployment.

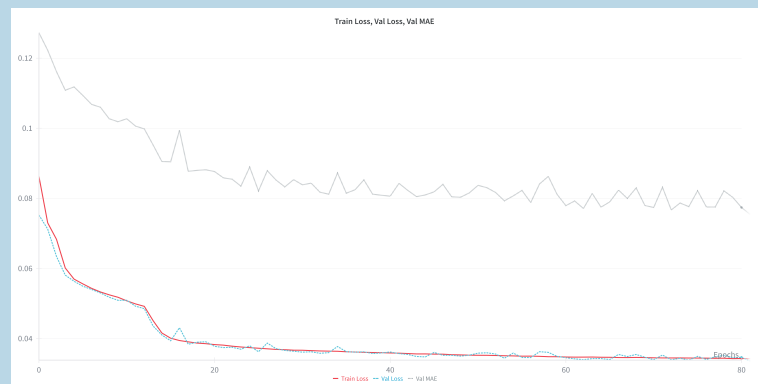
Optimization and training

MILP optimization: Fig(b) shows the a flow departing the network and Fig(a),(c),(d) shows arrival of flows. Link utilization is updated sequentially after every such routing.



Edge-based GNN Prediction: The GNN learns optimization scheme of MILP based routing. Key results:

- Mean Absolute Error (MAE): **0.0745** over all 74 edges.
- Trained on **1 million** MILP-derived routing paths.



This demonstrates that the GNN successfully learns the optimization policy and provides real-time performance with MILP-level accuracy.